

Is Pass@ k Shrinkage Real, and Do Its Fixes Compose?

A Pre-Registered, Human-Audited Study in Small Language Models

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DRAFT. Red numbers are pre-registered expectations, not results; violet text depends on the results; blue marks open items. Edit only as described in EDIT_GUIDE.md.

Abstract—Reinforcement learning with verifiable rewards (RLVR) raises the single-sample accuracy of language models, but whether it also shrinks the set of problems a model can solve at all is disputed: some studies find base models overtaking RLVR-trained ones at large k , others report expansion under longer training, and others trace the decline to how Pass@ k is measured. The evidence on all sides is automatic (Pass@ k , entropy, similarity metrics, LLM judges), so nobody has checked with people what RLVR actually loses; and fixes acting within a rollout group and across prompts have never been tested together. We address all three gaps on Qwen2.5-0.5B-Instruct trained with GRPO on MATH: a pre-registered 2×3 factorial crosses a macro factor (none vs. a topic-level budget on positive advantage mass) with a micro factor (none, the per-prompt SELF filter, or mean-preserving rarity redistribution over solution modes), with 3 seeds per cell, and a blinded human audit checks what is lost. Vanilla GRPO raises MATH-500 Pass@1 from 30.0% to 36.5% while Pass@32 falls from 71.5% to 66.5%. Annotators judge 57% of the base model’s successes on lost problems to be valid reasoning, against 85% on matched retained ones, so measured shrinkage mixes genuine loss with lost lucky hits. The macro-micro composite shrinks least, but the benefit of composing the two levels is bounded within $[-0.27, +0.63]$ pp per unit $\ln k$.^[R1] The human labels also support the solution-mode partitioner^[R17] (balanced accuracy 77%). The protocol runs on one 24 GB GPU and is released.

Index Terms—reinforcement learning with verifiable rewards, GRPO, Pass@ k , reasoning boundary, small language models, credit assignment, human evaluation, pre-registration

I. INTRODUCTION

Group Relative Policy Optimization (GRPO) [1] and related RLVR methods have become the standard way to improve mathematical reasoning in open language models. Whether they teach models to solve new problems, or only to solve solvable ones more reliably, is disputed. Yue et al. [2] found that base models overtake their RLVR-trained versions when many samples are drawn, a Pass@ k inversion since analysed as a shrinking reasoning boundary [3]–[6]. Others report the

opposite: prolonged training expands the boundary [7], and RLVR comes out ahead when only correct chains of thought are counted [8]; a two-stage account reconciles the two by training duration [9]. A third view questions the measurement: aggregate declines partly reflect how few rollouts each training problem receives [10], and Pass@ k at large k rewards lucky hits [11]. The question matters most for small language models (SLMs), whose usefulness rests on sampling-based coverage (majority voting, verifier reranking) rather than on one greedy answer.

The evidence on every side is automatic. Pass@ k cannot tell a valid solution from a lucky guess; CoT-Pass@ k hands that judgment to LLM judges, which accept corrupted reasoning almost as often as clean reasoning [12]; and approach-level diversity has been measured by an LLM judge calibrated on 80 human-labelled pairs, with no human judging RLVR outputs [13]. To our knowledge, no study has asked people what RLVR actually loses: whether the solutions that disappear are distinct strategies, and whether the problems that leave the boundary were ever solved by valid reasoning.

A third gap concerns the fixes. *Intra-group* methods give rare correct modes of one prompt more credit than common ones [14]–[19]; *cross-prompt* methods reweight whole problems or tasks [3], [10], [20], [21]. They act on different axes but have never been combined and tested for complementarity (ReCo combines two corrections, both within one prompt [15]). We address all three gaps in one pre-registered study:

- *Is shrinkage real at SLM scale?* A multi-seed measurement on Qwen2.5-0.5B-Instruct [22] trained on MATH [23], with problem-level boundary entries and exits and the number of rollouts per training problem [10].
- *What is lost, according to people?* A blinded human audit of (H1) whether the solution modes our partitioner finds are distinct strategies and (H2) whether the base model’s successes on problems lost to RLVR were valid reasoning, against difficulty-matched retained problems.

- *Do the fixes compose?* A 2×3 factorial crossing a topic-level budget on positive advantage mass (CB, macro) with no micro term, the per-prompt SELF filter, or intra-group mode-rarity redistribution (MEG), with seed-paired primary contrasts and Holm correction.
- **Evidence that at 0.5B measured shrinkage mixes genuine loss with lost lucky hits, and that the two levels are largely redundant: their interaction is bounded near zero and the effect sits in the micro term.**^[R2]

Code, logs, the pre-registration and the annotation protocol are released at [\[TODO: repository URL\]](#).

II. RELATED WORK

Is shrinkage real? Yue et al. [2] showed that RLVR-trained models are overtaken by their base models at large k . Nguyen et al. [3] attribute this to negative interference and winner-take-all reinforcement and propose SELF, which trains only on problems whose greedy answer is wrong; Wu and Xuan [5] argue that RLVR cannot leave the base model’s support, Zhou and Li [6] locate the contraction at the first decision of a solution, and Zhou [4] diagnoses the inversion. Against this, ProRL [7] reports expansion after prolonged, KL-controlled training, Wen et al. [8] report expansion under CoT-Pass@ k , and Yao et al. [9] reconcile the two with an exploitation stage that shrinks and an exploration stage that expands. Yuan et al. [10] relate the decline to few rollouts per problem and gate problems by success count (BBG), and Dragoi et al. [11] show that Pass@ k is biased toward lucky hits. Other responses change the optimizer or the search [24]–[26]. At SLM scale, GRPO dynamics [27], difficulty scaling [28] and reward granularity [29] have been studied, but not shrinkage.

Measuring what is lost. Lee et al. [13] show that embedding similarity tracks phrasing more than strategy, that word-bigram overlap tracks approach better, and that only 0.7% of GSM8K problems admit more than one approach against 33% on MATH; their RLVR results come from an LLM judge calibrated on 80 human-labelled pairs. Yang et al. [30] code the strategies of frontier models with AI coders and human adjudication, without RL. Taşaltı et al. [12] find that CoT-Pass@ k judges accept corrupted chains, and that Pass@ k exceeds CoT-Pass@ k by 19.7 points on Qwen2.5-generation solvers. Our audit puts people directly on RLVR outcomes.

Intra-group credit redistribution. Cue-GRPO [14] redistributes positive credit over a partition of the correct rollouts by rarity, leaving incorrect rollouts untouched and restoring the mean; we adopt its rule as our micro term. ReCo [15] reweights against distributional concentration at the response and token level; GCPO [16], DRA-GRPO [17], EDAS [18] and TreeAdv [19] reshape advantages using diversity or structure within the group.

Cross-prompt reweighting. MT-GRPO [21] reweights explicit tasks for worst-task reliability; CurveRL [20] reweights prompts by their pass-rate distribution; BBG gates problems by success count [10]; SELF selects by greedy failure [3]. CB differs in granularity (latent topics rather than tasks or single

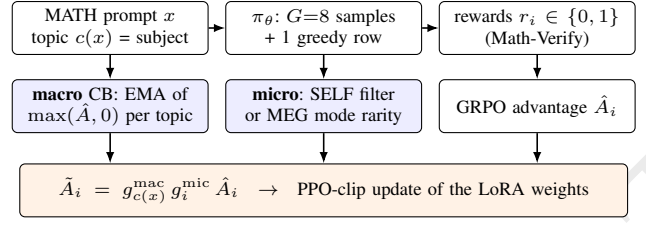


Fig. 1. Gated GRPO. The macro gate sees which topic a prompt belongs to and how much positive advantage that topic has absorbed in past steps; the micro gate sees the rollouts (and greedy answer) of the current group. Both scale the same advantage.

TABLE I
THE 2×3 FACTORIAL. COMPOSITES MULTIPLY THE MACRO AND MICRO GATES.

macro factor	micro factor		
	none	SELF filter	MEG redistribution
none	VANILLA	SELF	MEG
CB topic budget	CB	CB $\times$ SELF	CB $\times$ MEG (HEG-GRPO)

prompts), in signal (each topic’s share of positive advantage mass), and in mechanism (a stateful multiplicative gate). Our question is not which gate is best but whether a topic-level and an intra-group intervention compose.

III. METHOD

A. GRPO with gated advantages

For a prompt x the policy π_θ samples a group of G completions $y_1, \dots, y_G$ with binary rewards $r_i \in \{0, 1\}$ (answer correctness). GRPO [1] uses the group-normalized advantage

$$\hat{A}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \epsilon}. \quad (1)$$

Every condition in this study multiplies it by a non-negative gate weight, $\tilde{A}_i = g_i \hat{A}_i$ with $g_i = g_{c(x)}^{\text{mac}} g_i^{\text{mic}}$, and minimizes the PPO-clip loss [31], averaged over all T completion tokens in the batch [32]:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_i \sum_{t=1}^{|y_i|} \min(\rho_{i,t} \tilde{A}_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \tilde{A}_i), \quad (2)$$

where $\rho_{i,t}$ is the token probability ratio. Updates are on-policy and use no KL penalty. Vanilla GRPO is $g_i \equiv 1$. Fig. 1 summarizes the pipeline.

B. Macro gate: topic budget (CB)

Topics are the $C = 7$ MATH subjects. After each step, CB updates an exponential moving average of each topic’s spend, the mean positive advantage mass of its rollouts in that step, $s_c \leftarrow (1 - \eta) s_c + \eta \text{mean}_{i:c(x_i)=c} \max(\hat{A}_i, 0)$, with $\eta = 0.05$. Spend counts only positive mass, so corrective negative gradient never uses up a topic’s budget. With the overspend ratio $\rho_c = s_c / \frac{1}{C} \sum_{c'} s_{c'}$, the gate is

$$g_c^{\text{mac}} = \delta^{\max(0, \rho_c - \vartheta)}, \quad \vartheta = 1.5, \delta = 0.98, \quad (3)$$

TABLE II
PARTITIONER CALIBRATION ON BASE-MODEL ROLLOUTS (GROUPS WITH ≥ 4 CORRECT). RULE: MEDIAN MODES IN $[2, 4]$, CLOSEST TO 3; $\star$ = SELECTED.

MiniLM cosine τ	0.80	0.85	0.90	0.95	
median modes	1.0	1.0	1.5	2.5	
multi-mode groups (%)	5	10	35	70	
Word-bigram threshold	0.30	0.40	0.50	0.60	0.70
median modes	3.5	3.0*	2.0	1.5	1.0
multi-mode groups (%)	92	78	55	35	15

so topics that absorb more than ϑ times the average positive mass are throttled. Since $\rho_c \leq C$, $g^{\text{mac}} \in [\delta^{C-\vartheta}, 1]$. **[TODO: CB strength: with $C=7$ the gate never drops below 0.90; decide ϑ, δ from the pilot (EDIT_GUIDE §8)]**

C. Micro gates: SELF filter and mode-rarity redistribution (MEG)

SELF. Following [3], a prompt whose greedy answer is already correct gets $g_x^{\text{self}} = \lambda$ ($\lambda = 0$, a hard filter), otherwise 1. The greedy answer is decoded as one extra row of the same generation call (its logits are pinned to the argmax), so it costs one sequence per prompt. We implement the selection rule only, not SELF’s forward-KL term.

MEG. Let $\mathcal{C}$ be the N correct rollouts of a group, partitioned into solution modes, and let $|M(i)|$ be the size of the mode containing rollout i . Following Cue-GRPO [14],

$$w_i = \frac{N |M(i)|^{-\alpha}}{\sum_{j \in \mathcal{C}} |M(j)|^{-\alpha}} \quad (i \in \mathcal{C}), \quad w_i = 1 \quad (i \notin \mathcal{C}), \quad (4)$$

with $\alpha = 0.8$; weights are clipped to $[0.3, 3]$ and rescaled to mean 1 over $\mathcal{C}$. Because the mean is preserved, MEG changes *which* correct solutions receive credit, not how much; an earlier down-weight-only version was confounded with a lower learning rate. We call this term the mode-equalizing gate.

Mode partitioner. Modes come from agglomerative clustering of the correct completions, either on MiniLM [33] sentence embeddings [34] (cosine threshold) or on word-bigram Jaccard distance. Because embeddings track phrasing more than strategy [13], the partitioner is not hand-tuned: a pre-registered rule, applied to base-model rollouts before training, picks the candidate whose median number of modes per group lies in $[2, 4]$ and is closest to 3 (ties go to bigram). It selected **word-bigram Jaccard, threshold 0.4** (median 3.0 modes; 75% of groups with more than one mode; Table II). The rejected embedding candidates illustrate the risk: at the thresholds commonly used, **they place almost every group’s correct rollouts in a single mode**^[R15], which would turn MEG into a no-op.

A control condition reuses the composite gate with random mode labels (same number of modes per group).

D. Factorial design

Table I lists the six conditions. Composites multiply the two gates. Two exploratory extras complete the picture: CB $\times$ MEG

TABLE III
TRAINING AND EVALUATION SETTINGS, IDENTICAL FOR ALL CONDITIONS.

Model	Qwen2.5-0.5B-Instruct, bf16 weights
Adapters	LoRA $r=16$, $\alpha=32$, dropout 0, all projections
Optimizer	AdamW, lr 1×10^{-5} , cosine, 10% warm-up, grad-norm clip 1.0
Batch	16 prompts $\times G=8$ rollouts = 128 rollouts per step
Steps	400; MATH levels 1–5
Sampling	$T=0.7$, top- p 0.95, no top- k , $\leq 1,024$ new tokens
Loss	PPO-clip $\varepsilon=0.2$, on-policy, token-level mean, no KL
CB / SELF / MEG	$\eta=0.05$, $\vartheta=1.5$, $\delta=0.98$ / $\lambda=0$ / $\alpha=0.8$, clip $[0.3, 3]$
Evaluation	$n=32$ samples, $T=0.6$, top- p 0.95; MATH-500 (1,024 tokens), GSM8K 500-problem subset (512 tokens)
Seeds	3 per condition
Hardware	one NVIDIA A10G 24 GB (AWS g5.xlarge)

with random partitions, and a GRPO-compatible version of BBG’s success-count gate [10] (without its signed REINFORCE term for all-wrong groups, which group-normalized GRPO cannot express).

IV. EXPERIMENTAL SETUP

Training. We train Qwen2.5-0.5B-Instruct [22] with LoRA [35] adapters on the MATH training split [23] (7,496 problems with a boxed answer), prompted with the standard “reason step by step and put the final answer in `\boxed{\}`” instruction; Table III lists every setting. Rewards are binary correctness checked with Math-Verify [36] under a time limit, with a guarded symbolic fallback; a strict tag-format reward was dropped after it proved identically zero, hence inert under group normalization. The budget amounts to 0.85 epochs, or 6.8 rollouts per training problem. Seed s fixes the data order and LoRA initialization across conditions, so conditions are compared in seed-paired blocks.

Evaluation. Every model, including the base model, is scored on the same problems: all of MATH-500 [37] (in-distribution, primary; 1,024 tokens) and a fixed 500-problem subset of the GSM8K test set [38] (out-of-distribution; 512 tokens), with $n = 32$ samples per problem at temperature 0.6 and top- p 0.95. Pass@ k for $k \in \{1, 2, 4, 8, 16, 32\}$ uses the unbiased estimator [39] $1 - \binom{n-c}{k} / \binom{n}{k}$ for c correct samples. MATH-500 shares no problem statement with the training split; 7 of its problems (1.4%) are template siblings of training problems with a different question or numbers.

Outcome. The pre-registered primary outcome is the *shrinkage slope*: the least-squares slope of $\Delta(k) = \text{Pass}@k(\text{run}) - \text{Pass}@k(\text{base})$ against $\ln k$, reported in percentage points per unit $\ln k$. A negative slope means gains at small k turn into losses at large k ; higher means less shrinkage. We also count *boundary exits* (problems the base model solves at least once in 32 samples and the trained model never does) and *entries* (the reverse).

Pre-registered analysis. All contrasts are linear combinations of condition slopes, paired by seed. The two primary contrasts are P1, CB $\times$ MEG – CB $\times$ SELF (intra-group re-

distribution vs. per-prompt filtering, both under the budget), and P2, CB×MEG – MEG (does the macro level add to the micro level?). Secondary contrasts are the two macro×micro interactions (S1 for MEG, S2 for SELF), reported as 95% bounds rather than verdicts, and the macro main effect (S3); E1–E3 are exploratory. Each contrast gets a t -interval over seeds and a hierarchical bootstrap that resamples seeds and problems jointly (2,000 resamples). An effect is claimed only if its Holm-adjusted [40] bootstrap $p < 0.05$ and its bootstrap interval excludes zero. Diagnostics, logged for every condition: normalized mode entropy of the correct rollouts in each group, the mean gate weight on positive-advantage rollouts, and the share of prompts SELF selects. The pre-registration also fixes how outcomes are worded: a significant P1 or P2 means the composition helps beyond its parts; if neither is significant and the intervals are narrow, the result is a bounded null; if vanilla GRPO shows no shrinkage, that is itself reported and the contrasts are still given.

Human audit (pre-registered, secondary). Two annotators [TODO: background], blind to condition, group and partitioner output, label two sheets built by our release. *H1, partition validity:* 60 pairs of correct base-model solutions to the same training problem, half placed in one mode by the partitioner and half in different modes, labelled SAME or DIFFERENT approach under the approach-level definition of [13]. We report Cohen’s κ and the partitioner’s balanced accuracy against the annotators’ consensus, and read the modes as approach-level only if it reaches 70%. *H2, what is lost:* MATH-500 problems the base model solves at least once in 32 samples but vanilla GRPO never solves in a majority of seeds (*exited*), each matched to a retained problem with the closest base solve count. For one random correct base-model solution per problem, annotators judge the reasoning VALID or FLAWED (right answer, unsupported reasoning). Rates are compared with a two-sided Fisher exact test; the pre-registered reading is “mostly lucky hits” if the exited rate is below 50% and significantly below the retained rate, “genuine loss” if it is at least 50% and not significantly lower, and “mixed” otherwise.

Reproducibility. The pre-registration (committed [TODO: date + commit hash] before any main run), configs, per-step logs, sampled rollout groups for auditing the partitions, evaluation correctness matrices, and the script that generates every number in this paper are released at [TODO: repository URL].

V. RESULTS

A. Shrinkage at 0.5B

Vanilla GRPO reproduces the inversion at this scale^[R3]: Pass@1 rises from 30.0% to 36.5%, Pass@32 falls from 71.5% to 66.5% (Table IV), and the curves cross near $k \approx 6$ (Fig. 2, left). 24.7 problems leave the boundary while 8.3 enter it. Because aggregate shrinkage depends on how often each training problem is visited [10], we note that each training problem received 6.8 rollouts on average; the effect is therefore measured in the low-visit regime typical of SLM budgets.

TABLE IV
MATH-500 PASS@ k (%), MEAN OVER SEEDS), SHRINKAGE SLOPE (PP PER UNIT $\ln k$; HIGHER IS BETTER) AND BOUNDARY ENTRIES/EXITS (OF 500 PROBLEMS). ROWS BELOW THE RULE ARE EXPLORATORY EXTRAS.

Condition	P@1	P@4	P@16	P@32	Slope	In / Out
Base model	30.0	49.8	65.5	71.5	–	–
VANILLA	36.5	51.0	62.0	66.5	−3.41	8.3 / 24.7
SELF	35.4	50.9	62.8	67.6	−2.76	9.3 / 20.7
MEG	35.8	51.8	64.0	68.9	−2.50	10.0 / 18.3
CB	36.2	51.0	62.4	67.0	−3.17	8.7 / 22.3
CB×SELF	35.2	51.1	63.2	68.1	−2.56	9.0 / 19.7
CB×MEG	35.6	52.4	65.1	70.1	−2.09	10.7 / 15.3
CB×MEG, rand. modes	36.1	51.1	62.5	67.2	−3.08	8.7 / 22.0
BBG-style gate	35.0	51.1	63.4	68.4	−2.41	9.7 / 19.0

TABLE V
CONTRASTS ON THE MATH-500 SHRINKAGE SLOPE (PP PER UNIT $\ln k$), PAIRED BY SEED. CI: HIERARCHICAL BOOTSTRAP OVER SEEDS AND PROBLEMS. p : HOLM-ADJUSTED FOR P1–P2, UNADJUSTED OTHERWISE. GSM8K: ESTIMATE OF THE SAME CONTRAST OUT OF DISTRIBUTION.

Contrast	Est.	95% CI	p	GSM8K
P1 CB×MEG – CB×SELF	+0.47	[+0.02, +0.92]	0.085	+0.09
P2 CB×MEG – MEG	+0.41	[−0.04, +0.86]	0.085	+0.12
S1 CB×MEG interaction	+0.18	[−0.27, +0.63]	0.443	+0.03
S2 CB×SELF interaction	−0.03	[−0.48, +0.42]	0.901	+0.00
S3 CB main effect	+0.28	[−0.17, +0.73]	0.218	+0.10
E1 CB×MEG – Vanilla	+1.31	[+0.86, +1.76]	<0.001	+0.50
E2 CB×MEG – random modes	+0.99	[+0.54, +1.44]	<0.001	+0.38
E3 CB×MEG – BBG-style	+0.32	[−0.13, +0.77]	0.164	−0.00

B. Pre-registered contrasts

Every gate has a higher slope than vanilla (Table IV), and CB×MEG shrinks least^[R4] (−2.09 vs. −3.41). Neither primary contrast clears the pre-registered bar, however (Table V): P1 = +0.47 (CI [+0.02, +0.92], Holm p = 0.085) and P2 = +0.41 (CI [−0.04, +0.86], Holm p = 0.085) both point toward the composite, but their intervals include zero.^[R5] The CB×MEG interaction lies within [−0.27, +0.63] and the CB×SELF interaction within [−0.48, +0.42], so any super-additive benefit of composing the levels is at most a fraction of vanilla’s shrinkage slope; the macro main effect (S3 = +0.28) is small.^[R6] Following the pre-registration we report these as bounds, not as evidence of absence.

C. What each level does

Mode entropy starts at 0.46 (the same base model for every condition) and decays under vanilla GRPO, while both MEG conditions retain it^[R7] (Fig. 2, right; training means 0.31 for VANILLA vs. 0.43 for CB×MEG). The macro gate, in contrast, rarely binds^[R8]: the mean weight on positive-advantage rollouts is 0.98 under CB, against 0.55 under SELF, which filters 58% of prompts on average. Shrinkage is also uneven across the topics the macro gate acts on (Fig. 3): under vanilla GRPO, Pass@32 changes by −8.2 pp in Geometry but −2.2 pp in Prealgebra. CB flattens this profile without reducing the average loss, whereas both MEG conditions reduce it in every subject.^[R16]

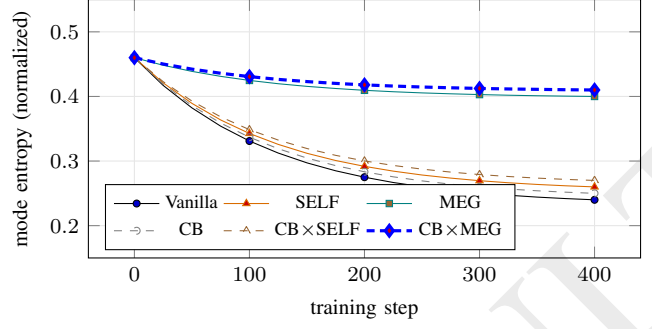
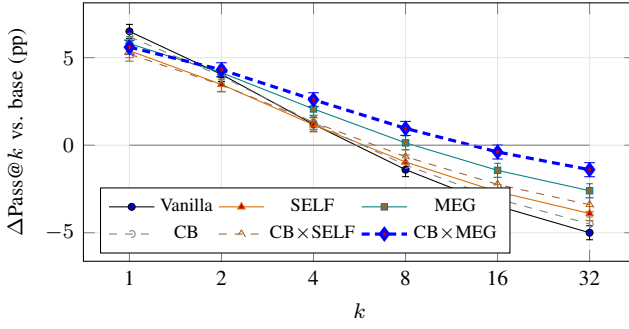


Fig. 2. Left: change in MATH-500 Pass@ k relative to the base model (mean $\pm$ s.d. over seeds); a curve that falls below zero at large k has shrunk the reasoning boundary. Right: normalized entropy of the solution-mode distribution among correct rollouts during training (mean over seeds).

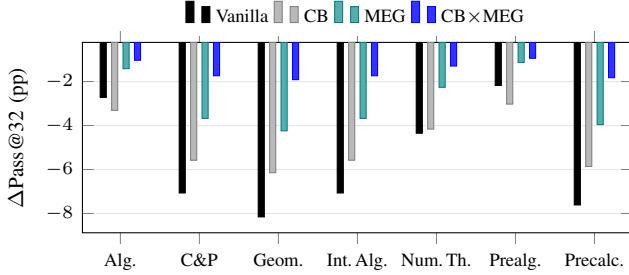


Fig. 3. Change in MATH-500 Pass@32 relative to the base model by subject, i.e. by the macro gate’s topics (mean over seeds).

Boundary exits fall^[R9] from 24.7 (VANILLA) to 15.3 (CB×MEG). Replacing the calibrated modes with random labels removes most of the gain, so the effect depends on the partition rather than on reweighting noise^[R10] ($E2 = +0.99$, CI $[+0.54, +1.44]$); the BBG-style gate ($E3 = +0.32$) lies between vanilla and the MEG conditions^[R11].

Out of distribution. On GSM8K the same ordering holds with smaller magnitudes^[R12]: slopes are -1.58 for VANILLA and -1.09 for CB×MEG ($P1 = +0.09$, CI $[-0.36, +0.54]$). Training on MATH thus does not trade in-distribution coverage for out-of-distribution coverage.^[R12]

D. What is lost: the human audit

Annotators agreed with $\kappa = 0.62$ on H1 and $\kappa = 0.68$ on H2. Against their consensus on 51 of 60 pairs, the partitioner reaches 78% accuracy (balanced 77%), above the pre-registered 70% bar, so MEG’s modes can be read as approaches rather than surface clusters.^[R17] Vanilla GRPO lost 61 problems in a majority of seeds. Of the base model’s successes on the 40 audited exited problems, 57% (CI $[42, 71]$) were valid reasoning, against 85% (CI $[71, 93]$) on matched retained problems (Fisher $p = 0.013$). By the pre-registered rule this is a *mixed* outcome: a sizeable share of what Pass@32 counts as lost was never solved by valid reasoning, but most of it was.^[R18]

VI. DISCUSSION AND LIMITATIONS

The three questions have connected answers at this scale. Shrinkage is real but overstated by Pass@ k : part of what

vanilla GRPO loses was reached by flawed reasoning, consistent both with the inversion literature [2], [3] and with the measurement critiques [11], [12]. The genuine part is concentrated within groups, where one correct mode crowds out others; an intervention at that level recovers most of it, and a topic-level budget adds little because topics are not where the gradient concentrates.^[R13] This holds for 0.5B models in a low-visit regime, far from the prolonged training in which expansion has been reported [7], [9]; the bounds in Table V are what a larger study would need to beat.

Limitations. (i) One model family at one size; a 1.5B replication was planned as optional **[TODO: state whether it was run]**. (ii) Three seeds bound the power for interactions, which is why they are reported as bounds. (iii) Solution modes are proxies: bigram and embedding partitions approximate strategy, and the random-partition control only shows that the partition matters, not that it equals strategy. (iv) SELF is implemented without its forward-KL term, and BBG without its signed loss for all-wrong groups. (v) LoRA rather than full fine-tuning, a single RL algorithm, and short training (6.8 rollouts per problem). (vi) The macro gate’s strength is fixed by the pre-registration; a stronger budget might behave differently. (vii) The human audit uses two annotators and 60+80 items on one model and one benchmark; H1 validates the partitioner on base-model rollouts only.

VII. CONCLUSION

We asked three questions about Pass@ k shrinkage in a small model: is it real, what does it lose according to people, and do topic-level and intra-group fixes compose? In a pre-registered study of Qwen2.5-0.5B-Instruct with a blinded human audit, shrinkage is real but partly a loss of lucky hits, and the composite of the two fixes shrinks least while the benefit of composing them is bounded near zero^[R14]. The protocol (seed-paired pre-registered contrasts, calibrated partitions with a random control, diagnostics for every condition, and a human audit of what is lost) runs on a single 24 GB GPU in 165 GPU-hours, and we release it so that larger studies can ask the same questions at scale.

ACKNOWLEDGMENT

[TODO: AWS credits (hackathon name), the two annotators, supervisor]. Total compute: 165 A10G GPU-hours (96 for training), about \$166.

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